

Robust model-based predictive controller for hybrid system via parametric programming

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Abstract

In this paper we present an algorithm for the design of robust model-based predictive control for hybrid system under uncertainty via parametric programming. The proposed min-max hybrid control scheme guarantees feasible plant operation for the maximum violation of uncertainty scenario. The key advantage of the proposed hybrid controller design is reduction in expensive, repetitive in nature on-line computations by providing the entire map of optimal robust control policy in given state space. The resulting piecewise affine optimal control law as a function of states can then be implemented on-line as a sequence of simple function evaluations. An example is presented to illustrate the details of the proposed robust hybrid parametric controller design.

Keywords: Hybrid systems, MPC, robust control, parametric programming, feasibility.

1. Introduction

Parametric programming is the state-of-the-art technology to find optimal solution of optimization problems under parametric uncertainty without exhaustively enumerating the entire parameter space, [1,2,3]. Application of parametric programming to the control of dynamical systems led to the development of off-line parametric controller, [4], also known as explicit model predictive control (MPC), [5,6]. The major advantages of explicit parametric controller is reduction in expensive, repetitive in nature online computations of MPC by performing “you solve only once” off-line computation, [3]. Thus, the forefront of “decision-to-policy” making parametric controller design is envisioned to encompass much wider range of applications including control of hybrid systems.

Modelling, optimization and control of hybrid systems, [7,8,9], is one of the most active areas of research in process systems engineering. Many practical engineering applications are inherently hybrid in nature that involve interactive combination of logic, dynamics and constraints; also known as mixed logical dynamical (MLD) systems, [10], or simply hybrid dynamical models, [11]. In the MPC framework this naturally leads to the mixed integer formulation of the MPC problem, where logical decisions are modeled as integer variables, [10,12].

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Additionally, many process systems are under the influence of uncertainties arising due to parameter variations and exogenous disturbances. Influence of uncertainty causes infeasible plant operation. Therefore one of the key control objectives for hybrid systems is to achieve robust stability and robust performance while guaranteeing economics and operational safety. However, the issue of robust controller design for hybrid systems under uncertainty is not completely addressed in the open literature. For recent review and progress in this area refer to [9]. Mayne and Rakovic, [13] have proposed on-line MPC for hybrid systems. In the current work, we apply the off-line optimization tools via parametric programming to design explicit MPC for constrained linear hybrid systems, which is robust in face of bounded input uncertainty, [6].

The next section 2 presents problem formulation for the multiparametric robust hybrid control (mpRHC). In section 3, Lyapunov based stability is achieved by using principles of linear matrix inequalities, [14], and robust feasibility is guaranteed by using the flexibility analysis theory of Pistikopoulos and Grossmann, [15]. Section 4, presents the min-max control problem. Finally, section 5 presents an example.

2. Hybrid System Model

2.1 System representation

Consider the following discrete multi-model dynamical system:

$$x(k+1) = \begin{cases} A_1x(k) + B_1u(k) + Gw(k) & \text{if } S_1x(k) + T_1u(k) \leq E_1 \\ A_2x(k) + B_2u(k) + Gw(k) & \text{if } S_2x(k) + T_2u(k) \leq E_2 \\ \vdots & \\ A_sx(k) + B_su(k) + Gw(k) & \text{if } S_sx(k) + T_su(k) \leq E_s \end{cases} \quad (1)$$

where $x(k) \in \mathbb{R}^n$, $u(k) \in \mathbb{R}^m$ and $w(k) \in \mathbb{R}^l$ are state, control and disturbance variables with $x(0) = x_0$ and corresponding system matrices $A_i \in \mathbb{R}^{n \times n}$, $B_i \in \mathbb{R}^{n \times m}$, $G \in \mathbb{R}^{n \times l}$; $\forall i = 1, \dots, s$. Furthermore, we enforce $x(k)$, $u(k)$ and $w(k)$ to be enclosed inside the bounded polyhedral sets i.e., $\forall k \geq 0$, $x(k) \in X$, $u(k) \in Y$, and $w(k) \in \Theta$ representing operating limitations. S_i, T_i , and E_i defines the convex polyhedra in the state space.

2.2 Reformulation to Mixed-Integer Form

Consider a binary variables $\delta_i(k) \in \{0,1\}$ corresponding to each of the i^{th} system dynamics, by defining the non-linear terms $x_i(k) = [A_ix(k) + B_iu(k)]\delta_i(k)$ system (1) can be reformulated as,

$$\begin{aligned} x(k+1) &= \sum_{i=1}^s z_i(k) + Gw(k) \\ z_i(k) &\leq M\delta_i(k); \quad z_i(k) \geq m\delta_i(k) \\ z_i(k) &\leq A_ix(k) + B_iu(k) - m(1 - \delta_i(k)); \quad z_i(k) \geq A_ix(k) + B_iu(k) - M(1 - \delta_i(k)) \\ E_i &\leq A_ix(k) + B_iu(k) - M^*(1 - \delta_i(k)) \end{aligned} \quad (2)$$

where $\sum_{i=1}^s \delta_i(k) = 1$, while $m = -M$, M^* are appropriately dimensioned large numbers.

2.3 Problem Formulation

The finite-horizon MPC problem for the hybrid system is given by,

$$\begin{aligned} \Phi_p(x(0)) = \min_{u(0)} & \left\{ \sum_{k=0}^{N-1} \|Qx(k)\|_p + \|Ru(k)\|_p \right\} + \|Px(N)\|_p \\ \text{s.t. } & x(k+1) = A_i x(k) + B_i u(k) + Gw(k) \text{ if } S_i x(k) + T_i u(k) \leq E_i \\ & x(k) \in X, u(k) \in Y, w(k) \in \Theta, x(N) \in O_\infty \subseteq X; \forall k \geq 0; \forall i = 1 \vee 2 \vee \dots s \end{aligned} \quad (3)$$

where $Q \succ 0$ and $R \succ 0$ are the weighting matrices for state and control while positive definite P is the stabilizing terminal cost for the prediction horizon N . The objective is defined over $p=1,2$ or ∞ based on l_1, l_2 or l_∞ performance criterion and $\vee$ is disjunction denoting logical “or” for $i=1, \dots, s$ systems. After the N^{th} time step we enforce the solution of constrained and unconstrained problem to coincide, [17,18], by defining O_∞ as the positive invariant set containing origin in its interior:

$$O_\infty \equiv \left\{ x(k) \in \mathfrak{X}^n, u(k) \in \mathfrak{U}^m \mid Kx(k) \in Y, \right. \\ \left. (A_i + B_i K)x(k) + Gw(k) \in X; \forall w(k) \in \Theta; \forall k \geq 0 \right\} \quad (4)$$

where K is the optimal feedback gain. Rewriting the system (2) in terms of constraint sets X, Y and substituting $x(k)$ into the objective function of equation (3) and can be reformulated as following multiparametric mixed integer quadratic program.

$$\begin{aligned} \Gamma(x(0)) = \min_{U, Z, D} & E_{w \in \Theta^N} [\Phi(U, Z, D, W, x(0))] \\ \text{s.t. } & g(z_i(k), \delta_i(k), u(k), w(k), x(0)) \leq 0 \\ & \sum_{i=1}^s \delta_i(k) = 1; \delta_i(k) \in \{0, 1\}; \forall i = 1 \vee \dots \vee s; \\ & x(0) \in X, U \in Y^N, w(k) \in \Theta, x(N) \in O_\infty \end{aligned} \quad (5)$$

where the column vector $Z = [[z_1(1), \dots, z_s(1)]^T, \dots, [z_1(N-1), \dots, z_s(N-1)]^T]^T$ and $D = [[\delta_1(1), \dots, \delta_s(1)]^T, \dots, [\delta_1(N-1), \dots, \delta_s(N-1)]^T]^T$ are the optimization vectors. $W = [[w_1(1), \dots, w_s(1)]^T, \dots, [w_1(N-1), \dots, w_s(N-1)]^T]^T$ is the expected disturbance vector and $U = [[u(0)^T, \dots, u(N-1)^T]^T$ while $x(0)$ are the current states treated as parameters.

3. Theoretical Developments

3.1 Stability and Terminal Cost for l_2 Criterion

Definition 3.1.1 Assuming pairs (A_i, B_i) are both stabilizable and detectable, system (A_i, B_i) is asymptotically stable if there exists quadratic Lyapunov function given by $V(\xi) = \xi^T P \xi > 0$.

Using this definition, we find $P \succ 0$ from the following theorem.

Theorem 1 (Lyapunov Stability) According to Lyapunov stability theorem, an open-loop system is stable if and only if $\forall i = 1, \dots, s; \exists P = P^T \succ 0$ such that $A_i^T P A_i - P < 0$ and

closed-loop system pairs (A_i, B_i) are stable if and only if $\forall i = 1, \dots, s; \exists P = P^T \succ 0$ such that $(A_i + B_i K)^T P (A_i + B_i K) - P < 0$. With $\alpha = P^{-1}$ and $\beta = K\alpha$ it is converted to LMI,

$$\begin{bmatrix} \alpha & (A_i + B_i K)^T \\ (A_i + B_i K) & \alpha \end{bmatrix} \succ 0.$$

After N^{th} time step control law $u(k) = Kx(k)$ with gain $K = \beta\alpha^{-1}$ is implemented.

3.3 Feasibility

Definition 4.3.1 The robust polytopic parametric predictive controller steers the plant into the feasible operating region for a specific range of uncertain variations.

According to the flexibility analysis theory of [15], maximum constraint violation defines the feasible operating region. This feasible region is depicted by the feasibility constraints, $\psi(U, Z, D, x(0)) \leq 0$ given by,

$$\psi(U, Z, D, x(0)) \leq 0 \Leftrightarrow \max_{W, j} \left\{ \begin{array}{l} g_i(U, Z, D, W, x(0)) \\ x(0) \in X, U \in Y^N, W \in \Theta^N, D \in \{0, 1\}^{Ns}; \forall j = 1, \dots, J \end{array} \right\} \quad (7)$$

Equation (7) can be solved by identifying critical uncertainty points for each maximization as, if $\frac{\partial g_j}{\partial w(k)} > 0 \Rightarrow w(k)^{cr} = w(k)^{ub}$ or if $\frac{\partial g_j}{\partial w(k)} < 0 \Rightarrow w(k)^{cr} = w(k)^{lb}$.

Thus, by substituting the sequence of critical uncertainty, $w(k)^{cr}$ in the constraints set $g(\cdot)$, a multiparametric linear program is formulated as,

$$\begin{aligned} \psi(U, Z, D, x(0)) &= \max_{W, j} \{g_i(U, Z, D, W, x(0))\} \\ &= \min_{\varepsilon} \left\{ \begin{array}{l} \varepsilon \geq g_i(U, Z, D, W, x(0)) \\ x(0) \in X, U \in Y^N, W \in \Theta^N, D \in \{0, 1\}^{Ns}; \forall j = 1, \dots, J \end{array} \right\} \end{aligned} \quad (8)$$

Equation (8) can then be solved using the formal comparison procedure of [1].

4. Design of mpRHC

The feasibility constraints (7) from section 3.3 are incorporated in problem (5) to obtain the following open-loop robust predictive control problem,

$$\begin{aligned} \Gamma(x(0)) &= \min_{U, Z, D} \min_{w \in \Theta^N} E[\Phi(U, Z, D, W, x(0))] \\ \text{s.t. } &g(z_i(k), \delta_i(k), u(k), w(k), x(0)) \leq 0 \\ &\sum_{i=1}^s \delta_i(k) = 1; \delta_i(k) \in \{0, 1\}; \forall i = 1 \vee \dots \vee s \\ &x(0) \in X, U \in Y^N, w(k) \in \Theta, x(N) \in O_\infty \\ &\min_{\varepsilon} \{\varepsilon \geq g_i(U, Z, D, W, x(0))\} \end{aligned} \quad (9)$$

This open-loop robust predictive control problem is a bi-level optimization problem. Note that the inner minimization problem is equivalent to equation (8), which can be solved separately resulting into a set of linear feasibility constraints $\psi(\cdot) \leq 0$. Substituting it into equation (9) results in following single-level optimization problem:

$$\begin{aligned}
\Gamma(x(0)) = \min_{U, Z, D} \max_{w \in \Theta^N} & E[\Phi(U, Z, D, W, x(0))] \\
s.t. & g(z_i(k), \delta_i(k), u(k), w(k), x(0)) \leq 0; \psi(U, Z, D, x(0)) \leq 0; \\
& \sum_{i=1}^s \delta_i(k) = 1; \delta_i(k) \in \{0, 1\}; \forall i = 1 \vee \dots \vee s; x(0) \in X, U \in Y^N, w(k) \in \Theta
\end{aligned} \tag{10}$$

Remark 4.1 The solution obtained in section 4. is obtained as a piecewise affine optimal robust parametric predictive control policy as a function of states $U(x(0))$ for the critical polyhedral regions in which plant operation is stable and feasible $\forall w(k)$.

5. Design Examples

Example 1: Consider the following dynamical system

$$x(k+1) = \begin{cases} 1.5x(k) + u(k) & \text{if } x(k) \geq 0 \\ 1.1x(k) + u(k) & \text{if } x(k) < 0 \end{cases}$$

$-10 \leq x(k) \leq 10; -1.2 \leq u(k) \leq 2.2$. with $Q=1, R=1$, and $N=2$, using Theorem 1, $P=0.0054, K=-1.35$. For l_∞ performance criterion, the open-loop computations are performed and the resulting piecewise affine optimal parametric predictive control profiles as a function of initial state are tabulated in Table 1.

Table 1: Open-loop parametric solution for example 1

CR #	CR	$u(k)$
1	$-10 = x(0) = -4.2$	$u(0) = 2.2$ $u(1) = 2.2$
2	$-4.2 = x(0) = -2.2$	$u(0) = 2.2$ $u(1) = -1.1 x(0) - 2.42$
3	$-2.2 = x(0) = 0$	$u(0) = -x(0)$ $u(1) = 0$
4	$0 = x(0) = 0.88$	$u(0) = -1.3636 x(0)$ $u(1) = 0$
5	$0.88 = x(0) = 1.4667$	$u(0) = -1.2$ $u(1) = -2.0455 x(0) + 1.8$
6	$1.4667 = x(0) = 5.9111$	$u(0) = -1.2$ $u(1) = -1.2$

6. Conclusion

This paper presents an explicit solution to the robust MPC for linear hybrid systems via parametric programming. A min-max based feasibility analysis is described to deal with the worst-case uncertainty. The controller performance guarantees system stability and feasible operation. The resulting controllers yield a piecewise affine control law which can be implemented on-line by simple function evaluations.

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